
HS 563: Theory and Methods of Policy Analysis

Course Code	HS 563
Course Name	Theory and Methods of Policy Analysis
L-T-P-C	3-0-0-3
Prerequisites	IC 210: Probability, Statistics and Random Processes; or, after instructors approval.
Intended for	3rd, 4th year B. Tech.; M.S., Ph.D.
Distribution	Elective for all B. Tech. students
Approval	4th Senate (May 2013)

Course Contents

- **Introduction to Policy Analysis** Policy research and analysis; policy analysis versus natural science; goals versus analysis; philosophical frameworks for analysis; motivations for taking policy analysis. [6 Lectures]
- **Discounting and CBA** Discounting; real/nominal discount rates; social discount rate; borrowing, leasing, depreciation, and taxes; cost-benefit analysis. [8 Lectures]
- **Decision Analysis and Uncertainty** Elements of policy (decision) problems; decisions, outcomes, and values; time value of money; nature and sources of uncertainty; structuring decisions: decision trees; risk profile and using decision trees to make choices; sensitivity analysis; multi-attribute decisions. [8 Lectures]
- **Modeling Uncertainty** Probability basics; subjective probability and biases; probability models (binomial, poisson, exponential, normal, and beta distributions); using data to construct distributions; Monte Carlo estimation. [8 Lectures]
- **Modeling Preferences** Evaluating risk attitudes; cost-effectiveness; risk analysis; and, value of life analysis. [6 Lectures]
- **Conflicting Objective, Utility Axioms, Paradoxes, and Policy Implications** Axioms for expected utility; paradoxes; conflicting objectives analyses; implications for policy analysis [6 Lectures]

Text Books

1. Clemen, **Making Hard Decisions**, Clemen and Reilly, Duxbury/Thomson.
2. Morgan, Henrion, and Small, **Uncertainty: A Guide to Dealing with Uncertainty in Quantitative Risk and Policy Analysis**, Cambridge University Press.

References

1. Sepulveda et al., **Schaum's Outlines of Engineering Economics**, McGraw-Hill, Note this is a study guide type book on the subject of engineering economics.
2. Campbell, **Benefit-Cost Analysis**, Campbell and Brown, Cambridge University Press.
3. Boardman, Greenberg, and Vining, **Cost Benefit Analysis: Concepts and Practice**, 3rd Edition, Prentice Hall.